JOHN EBENEZER

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with hands-on experience building full-stack, backend, desktop, and AI-powered applications using TypeScript, JavaScript, React, Next.js, Node.js, Express.js, PostgreSQL, and MongoDB. Experienced in designing reliable APIs, offline synchronization systems, secure data workflows, automated integrations, and AI-powered developer tools.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL, PL/pgSQL, Java, C++
- **Frontend:** React, Next.js, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, Server-Sent Events (SSE), Webhooks, Inngest
- **Databases:** PostgreSQL, Supabase, MongoDB, SQLite, MongoDB Atlas Vector Search
- **AI / LLM:** Google Gemini API, RAG, Vector Embeddings, Transformers
- **Security:** JWT, OAuth 2.0, RBAC, PostgreSQL RLS, AES-256-CBC, PBKDF2-HMAC-SHA256, HMAC-SHA256
- **Testing / DevOps:** Jest, Vitest, Supertest, Git, GitHub, GitHub Actions, Docker, Docker Compose, Sentry

EXPERIENCE

Web Developer Intern — Zippy Digital Solutions *Nov 2025 – Apr 2026*

- Engineered an Electron desktop POS application with React, Node.js, Express, and SQLite, owning **12 frontend modules, 4 backend services, 7 SQLite tables, and 10 API route groups** across UI, APIs, data, authentication, and synchronization.
- Designed a bidirectional event-driven synchronization engine using a SQLite-backed pending-event queue and UUID-based deduplication, replaying offline events after reconnection while a background worker pulled inbound events every 2 minutes.
- Secured cloud backup and synchronization with OAuth 2.0, AES-256-CBC, PBKDF2-HMAC-SHA256, OS keychain storage, retry handling, and scheduled Google Drive backups.
- Implemented runtime SQLite schema migration and offline event queuing, then packaged the application for Windows NSIS and Linux AppImage/deb distributions with rebuilt native dependencies.

PROJECTS

LeadVelox - B2B Lead Management & SLA Orchestration Platform [Github] [Live]

Next.js · React · TypeScript · Supabase PostgreSQL · Inngest · n8n

- Engineered atomic lead ingestion using PL/pgSQL and PostgreSQL advisory locks for concurrency-safe rate limiting and relational identity grouping for duplicate detection.
- Orchestrated resilient webhook workflows with n8n and Inngest using HMAC-SHA256 verification, idempotency tracking, 10-second timeouts, and exponential-backoff retries.
- Developed a timezone-aware SLA engine with four-tier automated escalation through scheduled background jobs; enforced RBAC and PostgreSQL RLS across three roles.

DevGrasp — AI-Powered GitHub Repository Assistant [Github] [Live]

React · TypeScript · Node.js · Express · MongoDB Atlas Vector Search · Gemini · RAG · SSE

- Engineered an asynchronous RAG pipeline with incremental SHA-based indexing and batched embeddings, reducing redundant repository re-indexing work by **90%+** for unchanged files.
- Implemented semantic code retrieval and context assembly with MongoDB Atlas Vector Search, achieving approximately **80–250 ms** vector retrieval latency.
- Implemented SSE-based AI response streaming with approximately **600 ms–1.2 s time-to-first-token** across codebase chat, onboarding, bug tracing, PR review, and technical-debt analysis.
- Containerized and automated DevGrasp with multi-stage Docker builds, health checks, non-root execution, Jest/Supertest integration tests, GitHub Actions CI/CD, and Sentry monitoring; kept the combined client/server footprint **under 210 MB** with typical CI execution of **2m15s–3m**.

EDUCATION

B.Tech, Software Product Engineering (Kalvium) — Kalasalingam Academy of Research and Education

2024 – 2028 (Expected)

ACHIEVEMENTS

- **Open Source Contributor — Medusa: Submitted PR #17151 fixing Medusa issue #16863, with unit and integration regression tests; unit tests, lint, and build passed.**
- Finalist at the national-level hackathon conducted by IIT Palakkad among the top 24 competing teams, for an **AI-powered examination fraud detection system** that flags suspicious candidate behavior during online assessments.
- Conducted an offline **n8n Workflow Automation Workshop** for **150+ students**, introducing workflow automation concepts and practical business process automation.